Leakage-restricted main-text baselines under three information budgets

LEAKAGE BARRIER: target lithofacies labels y_t are hidden until final scoring

Labelled source wells W_s

curves x_s + facies y_s

Unlabelled target well(s) W_t

curves x_t + measured depth d_t; no y_t

Select and disclose the admissible information budget

1 Inductive DG

Global-LGBM

Global curves + missingness masks
source-fitted statistics only

NO TARGET AGGREGATES

2 Post-drilling

Hybrid · target normalisation

Global + whole-well z-scores
optional relative depth

FULL TARGET INTERVAL

3 Causal-prefix

Hybrid-while · wireline check

Global + causal-prefix z-scores
no total depth / relative depth

PAST + CURRENT ONLY

Shared CPU LightGBM backbone

main-text feature variants | early stopping

Optional post-processing

centred depth smoothing — OFFLINE only

Unequal-budget references (not shown): Global-RF and appendix CORAL-MLP

Evaluation: P1 random-well | P2 quadrant hold-out | P3 six-well stress

Macro-F1 + coverage-aware fixed-tail screens (worst-well / W10)